

Session 6 :

Training Your Own AI Model

We explored how AI learns from data - datasets, features, training vs testing, generalization, and overfitting. Now it's time to apply what we've learned and train your own AI model using a real dataset from Kaggle.

Let's get started!



Objectives:

01

Understand Kaggle

Understand what Kaggle is and how developers find datasets

02

Train a Model

Train an image classification model using Google Teachable Machine

03

Test Your Model

Test your model on brand new images from the test_set folder

04

Compare Data Amounts

Compare how different amounts of data (1, 10, around 100 images) affect performance

05

Observe Epochs

Observe what happens when you train for more epochs

Recap : What We Learned in Session 5

Concept	What It Means
Dataset	A collection of examples used to teach AI
Features	The clues AI pays attention to (ears, face shape, fur)
Training Data	Practice examples AI learns from
Testing Data	New examples that check if AI really learned
Generalization	Being able to handle NEW situations – the goal
Overfitting	When AI memorizes instead of truly learning
Bias	When data is unfair or leaves groups out
Garbage In = Garbage Out	Bad data = bad AI, no matter what

📌 **Today's Experiments:** We'll see how some of these concepts show up when we train real models!

Warm-Up

Imagine you want to train an AI to recognize cats and dogs.

Think about these questions:



Take Photos Yourself?

Would you take all the pictures yourself?



Download a Dataset?

Or would you download a dataset from somewhere?



Where to Find Images?

Where would you get thousands of cat and dog pictures?

AI Enthusiasts usually use public datasets. One of the most popular places to find datasets is **Kaggle**. Today we'll use a dataset from Kaggle!

What is Kaggle?

Kaggle is an online platform used by AI enthusiasts, data scientists, machine learning engineers and anyone looking for real-world datasets.



Datasets

Like the one we're using today



Competitions

Machine learning competitions



Tools

For building AI models



Community

For learning and sharing

Our dataset comes from:

[kaggle.com - Cat and Dog Dataset](https://www.kaggle.com/datasets/pschmitt/cat-dog-dataset)



Our Dataset Structure

After downloading, the folder looks like this:

```
cat-and-dog-dataset/  
├── training_set/  ← Images for TRAINING  
│   ├── cats/     ← Many cat photos  
│   └── dogs/     ← Many dog photos  
└── test_set/     ← Images for TESTING  
    ├── cats/     ← NEW cat photos (AI has never seen these!)  
    └── dogs/     ← NEW dog photos (AI has never seen these!)
```

Folder	What's Inside	Purpose
training_set/cats/	Hundreds of cat photos	Teaching the AI what cats look like
training_set/dogs/	Hundreds of dog photos	Teaching the AI what dogs look like
test_set/cats/	Many NEW cat photos	Testing - AI has never seen these!
test_set/dogs/	Many NEW dog photos	Testing - AI has never seen these!

📌 **Important:** The images in test_set are brand new - the AI has never seen them during training. This helps us check if it really learned or just memorized.

Mini Activity : Build the Dataset

If we want to train an AI to recognize cats and dogs, what kinds of images should our dataset contain?

 **Cat Breeds**

 **Dog Breeds**

 **Angles**

 **Backgrounds**

 **Lighting**

 **Ages**

Mini Activity : Answers

Cat Breeds

Persian, Siamese, tabby

Dog Breeds

Golden Retriever, Poodle, Labrador

Angles

Front, side, looking away, from above

Backgrounds

Homes, parks, streets, grass

Lighting

Bright sun, dim rooms, shadows

Ages

Kittens, puppies, adults, seniors

 **Question:** What would happen if our dataset only had white cats and brown dogs? Discuss with your mentor.

Using Datasets to Train AI Models

To train an AI model, we follow a simple process:



Get a Dataset

Like our Kaggle cat and dog photos



Train the Model

Let the AI learn patterns from the data



Test the Model

Check if it works on brand new images



Improve the Model

Add more data or adjust training

📌 Remember what we learned in Session 2:

Data + Algorithm → Training → Model → Prediction

Today we'll follow this process with our cat and dog dataset!

Training an Image Classification Model

An **image classification model** is a type of machine learning model that identifies objects in pictures.

Examples:

- Recognizing animals (like our cats and dogs)
- Detecting handwritten numbers
- Identifying plants from photos
- Recognizing faces to unlock phones

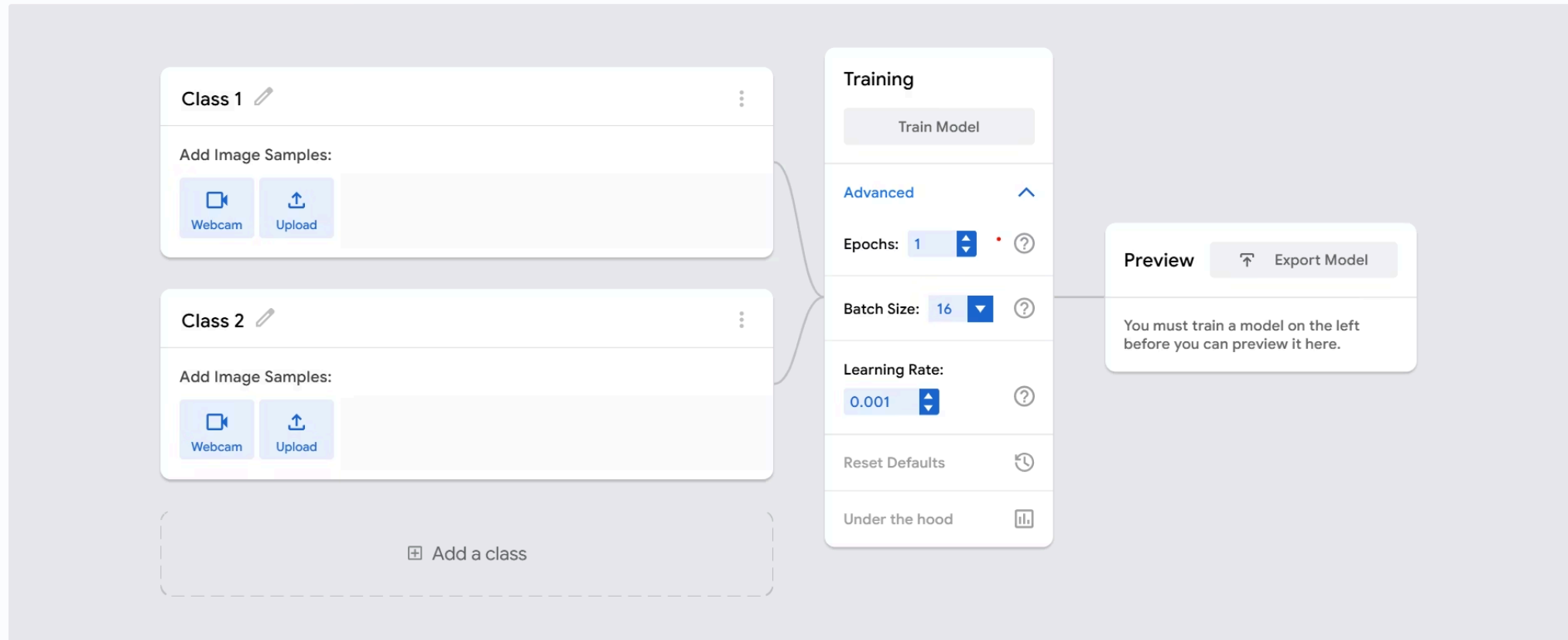
Models Learn By Analyzing:

- **Shapes** - round face, pointy ears, long snout
- **Colors** - fur color, eye color
- **Textures** - smooth fur, fluffy coat
- **Edges** - outlines of ears, eyes, mouth

After training, the model can predict which category a new image belongs to and give a **confidence score** showing how sure it is.

Google Teachable Machine

We'll use **Google Teachable Machine**, a free tool that lets you train simple AI models without coding. Here's what you'll see:



Understanding the Interface

Term	What It Does
Image Samples	Shows how many photos you've uploaded
Upload	Button to add images from your computer
Preview	Area to test your model with new images
Train Model	Button that starts the learning process
Advanced	Hidden settings for training
Epochs	How many times the AI sees all images
Output	Results showing confidence scores

Hands-On Activity : Cat vs Dog AI Model

We will run **4 simple experiments** using our Kaggle dataset. For testing, we'll use the same 4 images every time:

2 Cat Images

From test_set/cats/

2 Dog Images

From test_set/dogs/

Your mentor will tell you exactly which files to use (for example, "the first two cat images and first two dog images").

 **Why use the same test images every time?** So we can see how the model's performance changes as we change the training data and epochs.

Experiment 1

1 image per class, 1 epoch

Experiment 2

10 images per class, 1 epoch

Experiment 3

~100 images per class, 1 epoch

Experiment 4

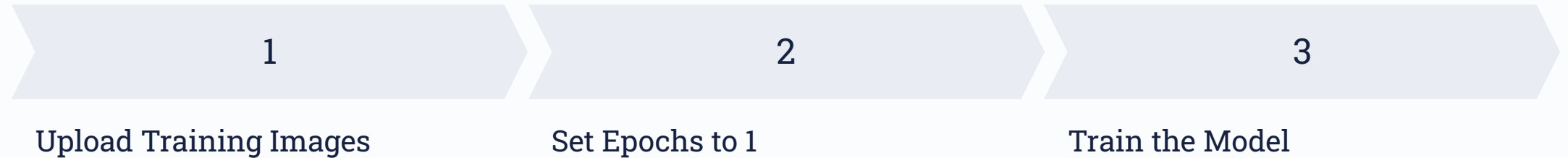
~100 images per class, 50 epochs



Experiment 1 : Very Little Data

1 image per class · 1 epoch

Prediction: What do you think will happen with only 1 cat photo and 1 dog photo?



Test Experiment 1

Now test your model on the 4 test images (provided by your mentor).

Image	Cat %	Dog %	Correct?
Test cat 1	___%	___%	
Test cat 2	___%	___%	
Test dog 1	___%	___%	
Test dog 2	___%	___%	

Total correct: __ out of 4

Observations

Question	Your Answer
Did it recognize the training images? (Try them!)	
Did it recognize new test images?	
Was confidence higher on training images or test images?	

 **Think about:** With only 1 image per class, did the AI learn patterns or just memorize?

Experiment 2 : A Bit More Data

10 images per class · 1 epoch

Prediction: What do you think will happen with 10 images per class?



Test Experiment 2

Test on the same 4 test images again.

Image	Cat %	Dog %	Correct?	Better than Exp 1?
Test cat 1	___%	___%		
Test cat 2	___%	___%		
Test dog 1	___%	___%		
Test dog 2	___%	___%		

Total correct: __ out of 4

Compare Results

	Experiment 1 (1 image)	Experiment 2 (10 images)
Correct out of 4	/4	/4

 What do you notice? Did more data help?

Experiment 3: More Data

~100 images per class · 1 epoch

Prediction: What do you think will happen with around 100 images per class?



Test Experiment 3

Test on the same 4 test images.

Image	Cat %	Dog %	Correct?	Compare to Exp 2
Test cat 1	___%	___%		
Test cat 2	___%	___%		
Test dog 1	___%	___%		
Test dog 2	___%	___%		

Total correct: __ out of 4

Compare Results

	Exp 1 (1)	Exp 2 (10)	Exp 3 (~100)
Correct out of 4	/4	/4	/4

 What do you notice? Did more data help? Why do you think that happened?

Experiment 4: More Training

~100 images per class · 50 epochs

Prediction: What do you think will happen if we train for much longer on the same images?



Test Experiment 4

Test on the same 4 test images one last time.

Image	Cat %	Dog %	Correct?	Compare to Exp 3
Test cat 1	___%	___%		Better? Worse?
Test cat 2	___%	___%		Better? Worse?
Test dog 1	___%	___%		Better? Worse?
Test dog 2	___%	___%		Better? Worse?

Total correct: __ out of 4

What Happened? Check One:

Got **BETTER** than Experiment 3

More training helped the model generalize even further.

Got **WORSE** than Experiment 3

The model may have started to overfit.

Stayed about the **SAME**

The model may have already learned all it could from the data.

What do you think caused this?

- ❏ If it got worse, this could be **overfitting** - when AI memorizes instead of learning. If it stayed the same, maybe it learned all it could from the data.

Compare All 4 Experiments

Experiment	Images per Class	Epochs	Correct out of 4
1	1	1	/4
2	10	1	/4
3	around 100	1	/4
4	around 100	50	/4

Which experiment worked best? Experiment ____

Key Takeaways

With very few images (1) - The AI could only memorize, not learn real patterns

With more images (10) - Performance improved a bit

With many images (around 100) - Performance improved a lot

With too much training (50 epochs) - Sometimes helps, sometimes causes overfitting



Real-World AI Uses the Same Process

The process you followed today - getting data, training, testing, improving - is exactly what professional AI engineers do, just at a much larger scale.

1M+

Images

ImageNet, a famous dataset, has over 1 million labeled images used to train real AI models

100+

Epochs

Professional models are often trained for hundreds of epochs with careful monitoring

95%+

Accuracy

State-of-the-art image classifiers can achieve over 95% accuracy on test data

① You've taken your first real step into the world of machine learning. The concepts you practiced today are the same foundations used by AI engineers at companies like Google, Meta, and OpenAI.

Summary : What We Learned Today

Kaggle

A website where data scientists share datasets

training_set

Folder with dataset used to teach the AI

test_set

Folder with brand new dataset to test real learning

Image Classification

AI that looks at pictures and decides what they show

Epochs

How many times the AI studies all the training images

Confidence Score

A percentage showing how sure the AI is about its prediction

Limited Data

Not enough examples to learn properly

Generalization

Being able to handle new situations

Overfitting

When AI memorizes instead of truly learning

Homework

Find a dataset online (any topic - images, text, sports, music, weather, anything interesting!)

Some Places to Look:

- Kaggle (explore their datasets section)
- Government data websites
- Sports statistics websites
- Any public dataset you can find

Answer These Questions:

1

What type of data does the dataset contain?
(Images? Numbers? Text? Audio?)

2

What problem could this dataset help solve?
(Example: "This dataset of house prices could help predict what a house is worth.")

3

What challenges might appear when training a model using this dataset?

Next Session Preview

Session 7: What is Generative AI?

You've trained AI models that **recognize** things (cats and dogs). But did you know AI can also **create** things?

Next session, we'll begin a new module: **Generative AI** - AI that can create new content like:



Text

Stories, poems, essays



Images

Art, photos



Audio

Voice, Sound



And More!

The possibilities are endless

Get ready to explore AI that creates!